



Colostrum fat content modulates glucose homeostasis and lipid metabolism in neonatal calves exposed to cold temperatures

Hannes Erkinger,¹ Maximilian A. E. von Riedheim,¹ Oguz Calisici,² and Josef J. Gross^{1*}

¹Veterinary Physiology, Vetsuisse Faculty, University of Bern, 3012 Bern, Switzerland

²Phytobiotics Futterzusatzstoffe GmbH, 65343 Eltville, Germany

ABSTRACT

Thermoregulation in neonatal calves is essential to maintain body temperature. Especially during the cold season, a successful metabolic adaptation is required to avoid health and production losses. Colostrum and transition milk are characterized by an elevated fat content compared with mature milk. Although colostrum fat content is known to vary considerably among cows, its specific effect on postnatal metabolic adaptation in calves under cold conditions has not been well characterized. The objective of the present study was to investigate the impact of a different fat content in colostrum and transition milk on metabolic adaptation in newborn calves during the cold season. Twelve calves ($n = 9$ Holstein, $n = 3$ Holstein $\times$ Limousin crossbred; birth weight 41.3 ± 1.8 kg) born to cows without dystocia were enrolled immediately after birth and studied until d 21 postpartum (p.p.). The experimental period lasted from February to March 2025. Based on sex, birth weight, and breed, calves were assigned to either a group receiving 2.5 L full-fat colostrum (FF; $n = 6$; 4.81% fat) or low-fat colostrum (LF; $n = 6$; 0.58% fat) at 4 and 12 h p.p. Both FF and LF derived from the same pool of colostrum and differed only in fat content. On d 2 and 3 p.p., FF and LF received transition milk (300 mL colostrum of different fat content mixed with 2.7 L bulk tank milk). From d 4 onwards, all calves were fed identically with bulk tank milk. Blood was sampled from d 2 to 7 and analyzed for various endocrine and metabolic parameters (e.g., glucose [GLU], insulin, cortisol, glucagon, nonesterified fatty acids [NEFA], triglycerides [TG], total cholesterol [TC], high-density lipoprotein cholesterol [HDL-C], low-density lipoprotein cholesterol [LDL-C], very low-density lipoprotein cholesterol [VLDL-C], aspartate aminotransferase [ASAT], gamma-glutamyl transferase [GGT], triiodothyronine, thyroxine, total protein [TP],

and IgG). Plasma GLU concentration increased during the first days after parturition in both LF and FF, with greater concentrations in FF during the first 4 d p.p. Serum concentrations of TG, TC, HDL-C, and VLDL-C were greater in FF until d 4 p.p., and TC and LDL-C were greater in FF compared with LF from d 5 to 21. No differences were observed for serum IgG, insulin, NEFA, and TP. Cortisol and glucagon concentrations as well as activity of GGT were elevated in LF compared with FF until d 4 of age. In conclusion, the present study showed that the fat content of colostrum and transition milk affected metabolic adaptation in neonatal calves during the first days of life. The overall lower energy availability in LF was reflected by changes of parameters related to glucose and lipid metabolism. Further studies investigating the impact of colostrum fat content under different environmental conditions are warranted.

Key words: colostrum, fat, energy metabolism, cold season

INTRODUCTION

Thermoregulation in neonatal calves is essential to maintain body temperature and to adapt to extrauterine environmental conditions (Nonnecke et al., 2009; Hammon et al., 2012; Silva et al., 2021). Especially during the cold season, successful adaptation is required to avoid health and production losses. Therefore, the timely supply with energy-rich compounds fueling thermoregulation by increasing calves' metabolic heat production through feeding is of utmost importance. Colostrum fat content varies considerably within and between mammalian species and is commonly greater in colostrum compared with mature milk (Bigler et al., 2022). Consequently, the fat content of colostrum represents a crucial energy source for the neonate (Odle, 1997). The present study aimed to provide new data on varying fat concentrations in colostrum and transition milk and their effects on neonatal calf metabolism. Specifically, we tested the hypothesis that different fat concentrations in ingested colostrum affect plasma glucose and lipoprotein concentrations, thyroid

Received December 16, 2025.

Accepted March 8, 2026.

*Corresponding author: josef.gross@unibe.ch

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

hormone levels, and other metabolic parameters. Previous studies in humans, pigs, and rats demonstrated the impact of dietary fat on gluconeogenesis and β -oxidation (Ferré et al., 1986; Girard, 1986; Lepine et al., 1989). Hammon and Blum (1998) reported increased plasma glucose concentrations in calves fed colostrum and milk replacer differing in fat content. However, more detailed investigations in neonatal calves' lipid and glucose metabolism, particularly when exposed to low temperatures during the cold season, are lacking. For neonatal calves, a lower critical temperature range between 8°C and 13°C has been reported in literature (Young, 1981; Litherland et al., 2014). Below that temperature, thermogenic demands increase, leading to elevated rates of heat production, mobilization of energy reserves, and altered endocrine responses such as changes in thyroid hormone activity, cortisol secretion, and glucose turnover (Polk, 1988; Bienboire-Frosini et al., 2023; Su et al., 2023). These physiological adaptations require adequate energy supply, particularly from dietary fat, to maintain normothermia and support early postnatal development. Therefore, colostrum fat content may be of particular relevance during the cold season, when metabolic demands are increased. Therefore, we hypothesized that reduced fat content in colostrum limits early energy availability and thereby compromises metabolic adaptation in neonatal calves exposed to cold ambient temperatures. To test this hypothesis, we investigated the metabolic responses of newborn calves receiving colostrum and transition milk with either normal or markedly reduced fat content for 3 d during the first 3 wk of life in winter.

MATERIALS AND METHODS

The animal experiment was carried out in accordance with the Swiss animal protection and welfare legislation and approved by the cantonal animal experimentation commission of the Canton of Fribourg (Switzerland, approval number 2024-23-FR).

Animals, Husbandry, and Feeding

The experimental period lasted from February to March 2025. Twelve clinically healthy Holstein and Holstein $\times$ Limousin crossbred calves (8 female, 4 male, birth weight 41.3 ± 1.8 kg) from the Agroscope Research Station in Posieux, Switzerland, were enrolled in the study at parturition until d 21 of life. Exclusion criteria were febrile systemic illness, dystocia, and twin calving. Calving was monitored by camera and barn staff. Calves were separated from their dams to avoid suckling and transferred into individual boxes with straw bedding no later than 30 min after parturition. After the first blood sampling at 4 h postpartum (**p.p.**), animals were housed

individually in outdoor hutches until d 7 and thereafter in pairs until the end of the study at d 21 p.p.

Based on birth weight and sex, calves were assigned to 1 of 2 treatment groups receiving either pooled full-fat (**FF**; $n = 6$; 4 Holstein/2 Holstein $\times$ Limousin crossbred, 3 female/3 male) colostrum (4.81% fat) or pooled low-fat (**LF**; $n = 6$; 4 Holstein/2 Holstein $\times$ Limousin crossbred, 5 female/1 male) colostrum (0.58% fat). Only 2 persons that were not involved in animal care and laboratory analyses were aware of group allocation and blinding of treatments. Each calf received 2.5 L of respective colostrum (FF or LF) at 4 and 12 h of age. Thereafter, calves were fed twice daily (0600 and 1600 h) with 3 L of milk. On d 2 and 3, the respective pooled colostrum (300 mL) was mixed with bulk tank milk (2.7 L) to simulate transition milk feeding. From d 4 onwards, all calves received the same bulk tank milk with added carbohydrates (lactose + maltodextrin 40 g/L; Multi-Lac Aufzucht, Multiforsa, Switzerland) according to the rearing protocol of the experimental station. Colostrum was collected from commercial dairy farms, pooled, and frozen in batches of 1,500 kg upon further processing. One batch of colostrum was thawed and split across 10 L containers. Half of the thawed colostrum was defatted via centrifugation at $10,000 \times g$ and 40°C for 3 min (Milchzentrifuge 2.0, MILKYSKY GmbH). Approximately 50 mL of colostrum, transition milk, and bulk tank milk were collected twice weekly, pooled, and contents of fat, protein, and lactose were determined by infrared spectroscopy (Süsslab, Zollikofen, Switzerland). The IgG concentration in colostrum, transition milk, and bulk tank milk was determined by ELISA (Bethyl Laboratories Inc., TX; bovine reference serum: reference [ref.] RS10-103 bovine reference serum, IgG antibodies, ref. A10-118A and ref. A10-118P). Energy content of colostrum, transition milk, and milk was calculated using a component-based gross energy equation (Sjaunja et al., 1990). Gross composition of colostrum, transition milk, and bulk tank milk is shown in Table 1. Except for the difference in fat and energy contents in colostrum and transition milk between FF and LF, contents of IgG, protein, and lactose did not differ. Water, hay, and concentrate were offered ad libitum.

Animal Data Recording

A health check including fecal and naval scoring (Kertz and Chester-Jones, 2004; McGuirk, 2008; McGuirk and Peek, 2014), respiratory and heart rate, and rectal temperature was done daily until d 7 and on d 14 and 21. Individual milk intake and residuals were recorded daily, and BW was measured at birth and on d 4, 7, 10, 14, and 21. Ambient temperature was recorded via temperature loggers placed near to the calf hutches.

Table 1. Nutrient composition of and energy intake via colostrum, transition, and bulk tank milk (mean values $\pm$ SD)¹

Item	FF colostrum (n = 6) ²	LF colostrum (n = 6) ²	FF transition milk (n = 3) ²	LF transition milk (n = 2) ²	Bulk tank milk (n = 9) ²
Fat (%)	4.81 $\pm$ 0.42	0.58 $\pm$ 0.17	5.17 $\pm$ 0.62	3.80 $\pm$ 0.49	4.51 $\pm$ 0.28
Protein (g/100 g)	10.32 $\pm$ 0.31	9.64 $\pm$ 0.78	4.19 $\pm$ 0.14	4.49 $\pm$ 0.13	4.23 $\pm$ 0.10
Lactose (g/100 g)	3.26 $\pm$ 0.25	3.18 $\pm$ 0.47	4.63 $\pm$ 0.06	4.58 $\pm$ 0.01	6.98 $\pm$ 0.11
IgG (mg/mL)	35.8 $\pm$ 1.7	31.9 $\pm$ 2.7	2.7 $\pm$ 1.8	2.2 $\pm$ 0.3	ND
Energy intake (MJ/d)	24.50	15.50	22.68	19.92	23.58

¹FF = full-fat; LF = low-fat; ND = not determined.²Numbers refer to the number of samples analyzed for gross composition and IgG content.

Blood Sampling and Analysis

Blood was collected using evacuated tubes (1 tube 9 mL for serum and 1 tube 9 mL for EDTA plasma harvest; Greiner Bio-One GmbH, Germany) via puncture of the jugular vein at 4 (before colostrum feeding), 12, and 24 h p.p. and once daily at d 2, 3, 4, 5, 7, 14, and 21 before morning feeding. Blood samples for EDTA plasma were placed on wet ice immediately after sampling, whereas serum samples were allowed to clot for 2 h at room temperature and were then stored at 6°C until centrifugation (1,500 $\times$ g, 4°C for 15 min). Serum and plasma samples were stored in 1.5 mL snap vial tubes at -80°C until analysis.

Plasma glucose (GLU) concentration was measured in plasma using an enzymatic assay (ref. 08057800190, Roche). Serum concentrations of total cholesterol (TC; ref. 08057443190, Roche), total protein (TP; ref. 08058652190, Roche), urea (ref. 08058806190, Roche), triglycerides (TG; ref. 08058687190, Roche), free fatty acids (NEFA; ref. FA115, Randox), phospholipids (PL; ref. 296-63801, Wako), and high-density lipoprotein cholesterol (HDL-C; ref. 08057877190, Roche) and activities of aspartate aminotransferase (ASAT; ref. 08104719190, Roche) and gamma-glutamyl transferase (GGT; ref. 08057796190, Roche) were determined with an autoanalyzer (Cobas Pure c303, Roche Diagnostics International Ltd., Switzerland). Concentrations of low-density lipoprotein cholesterol (LDL-C) and very-low-density lipoprotein cholesterol (VLDL-C) were calculated using equations as described earlier by Friedewald et al. (1972) and Kessler et al. (2014).

Plasma insulin and cortisol concentrations were measured using commercial kits from Beckman Coulter Life Sciences, Switzerland (IRMA insulin, ref. 3210, and RIA cortisol, ref. C23117). The concentrations of triiodothyronine (T₃) and thyroxine (T₄) were measured in blood plasma by RIA (Beckman Coulter Life Sciences, Switzerland; T₃: ref. IM1699 and T₄: IM1447). Serum glucagon was measured via RIA (EMD Millipore Corporation, ref. GL-32K). Immunoglobulin G in serum was analyzed as described above for milk samples.

Statistical Analysis

Statistical analysis was performed with SAS (version 9.4, SAS Institute Inc., Cary, NC) and SigmaPlot (version 16, Systat Software Inc.). Data are presented as mean values $\pm$ SEM. A power analysis (PROC POWER) with plasma glucose concentration as primary outcome was conducted to confirm that the chosen sample size provided a statistical power of >0.8 for detecting biologically relevant differences in the measured blood metabolites. A mixed model with group (FF, LF), time, and the interaction group $\times$ time as fixed effects and the individual calf treated as the repeated experimental unit was used to evaluate effects of FF or LF feeding on investigated parameters. Time was set as repeated factor and different covariance structures (variance components [VC], compound symmetry [CS], and unstructured) were tested. The covariance structure providing the best model fit based on Akaike information criterion and numerical stability was retained; in most cases, the VC structure resulted in the most stable estimates, whereas CS was used as a sensitivity check for key parameters to ensure robustness of results. Because feeding treatments differed only from 4 h p.p. to d 4 and were identical thereafter, analyses were performed separately for these periods, 4 h p.p. to d 4 and d 5 to d 21, respectively. Differences between FF and LF were identified using a Tukey's post hoc test ($P < 0.05$). Tendencies toward significance were assumed at $0.05 \leq P < 0.10$.

RESULTS

Environmental Temperature, Energy Intake, BW, and Vital Parameters

Ambient temperature tended to increase with advancing season. Average ambient temperature was $3.5 \pm 0.7^\circ\text{C}$ during the first 4 d of life (range -0.8°C to $+8.8^\circ\text{C}$; Figure 1). From d 5 to 21 p.p., calves were exposed to temperatures ranging from -0.8°C to $+12.4^\circ\text{C}$. Energy intake of calves is shown in Table 1. Due to the greater nutrient content, overall energy intake was greater in FF

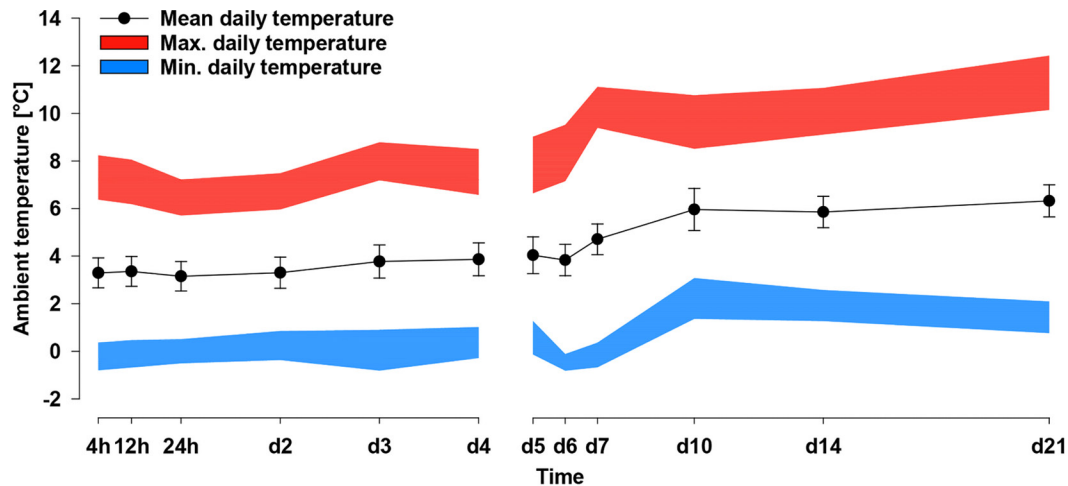


Figure 1. Ambient temperature during the experimental period. The graphic shows the mean, minimum (min.), and maximum (max.) temperatures. All data are presented as mean values $\pm$ SEM).

compared with LF ($P < 0.05$). Body weight increased in all calves throughout the experimental period (FF: 42.8 ± 2.4 kg to 62.5 ± 3.8 kg; LF: 39.8 ± 2.4 kg to 60.2 ± 3.3 kg). Body weight at d 1 did not differ between groups ($P = 0.49$). Weight gain did not differ between groups ($P = 0.61$). Vital parameters such as respiration rate, heart rate, and rectal temperature were not affected by treatment at any time and remained within normal physiological ranges throughout the study period (data not shown).

Plasma Metabolites Related to Glucose and Lipid Metabolism

During the period from 4 h p.p. to d 4, glucose concentration in plasma was greater in FF compared with LF ($P < 0.05$; Figure 2A). Although glucose concentration in LF remained at a similar level until 12 h p.p. (4.39 ± 0.32 mmol/L; $P = 0.83$) and increased thereafter until d 3 p.p., a continuous and steeper increase was observed in FF from 4 h p.p. up to 24 h p.p. (6.93 ± 0.33 mmol/L; $P < 0.05$). From 24 h to d 4, glucose concentrations remained constant in both the FF ($P = 0.84$) and LF groups ($P = 0.96$). From d 5 onwards, glucose concentration did not differ between groups ($P = 0.69$). Plasma NEFA concentrations showed a sharp decline in both groups between 4 h and 12 h p.p. ($P < 0.05$; Figure 2B). After a nadir at 12 h in LF, NEFA concentration slightly increased again and remained stable thereafter. No group differences were observed in NEFA concentration during the entire experimental period ($P > 0.05$; Figure 2B).

Serum TG concentrations in both LF and FF increased after birth but to a lower rate in LF compared with FF

(Figure 3A). Serum TG concentrations were greater in FF compared with LF during the first 4 d of life ($P = 0.02$) but did not differ among groups between d 5 and 21 p.p. ($P = 0.66$; Figure 3A). Phospholipid concentrations increased continuously throughout the experiment in both groups and were greater in FF compared with LF (4 h–d 4 p.p.: $P < 0.0001$; d 5–21 p.p.: $P = 0.09$; Figure 3B). Similar to PL concentration, total cholesterol and HDL-C in serum increased after birth and were greater in FF compared with LF throughout the entire observation period ($P < 0.05$; Figures 3C and 3D). The LDL-C concentrations did not differ between FF and LF until d 4 ($P = 0.53$; data not shown) but tended to be elevated in FF compared with LF during d 5 to 21 p.p. ($P = 0.07$). Similar to TG concentrations, concentrations of VLDL-C were greater in FF compared with LF until d 4 ($P < 0.05$; data not shown) but did not differ between groups thereafter ($P = 0.66$).

Total Protein, Urea, and IgG Concentrations

Total protein concentration increased significantly until 24 h p.p. and remained at a constant level until d 21 (Figure 4A). No difference in TP concentration between FF and LF was observed throughout the observation period ($P > 0.05$). Similar to the changes of TP, IgG concentration sharply increased after parturition, reaching a peak at 24 h p.p. (Figure 4B). Thereafter, IgG concentration declined slowly until the end of the study. No differences between FF and LF were observed. Urea concentration did not differ between LF and FF (4 h to d 4 p.p.: $P = 0.44$; d 5 to d 21 p.p.: $P = 0.89$; data not shown).

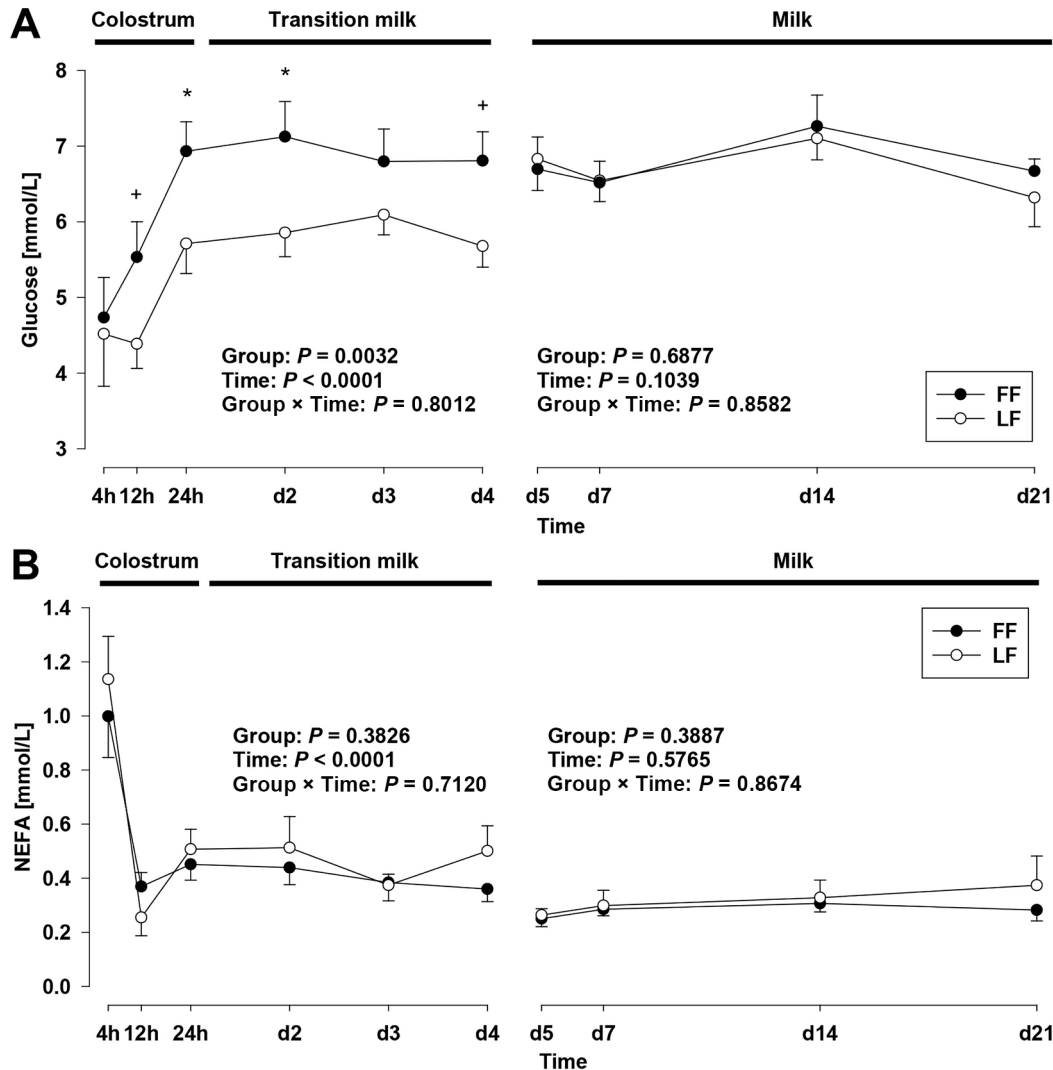


Figure 2. Concentrations of glucose (A) and nonesterified fatty acids (NEFA, B) in blood of calves during the experiment. The observation periods for FF ($n = 6$) and LF ($n = 6$) were divided into a colostrum phase (4 to 24 h), a transition milk phase (d 2 to 4), and a bulk tank milk phase (d 5–21). Data are presented as mean values $\pm$ SEM. Significant differences ($P < 0.05$) between groups are indicated by *, and tendencies toward significance ($0.05 \leq P < 0.10$) are marked by +.

Endocrine Parameters

Throughout the entire study period, no significant differences in insulin concentrations in plasma were observed between FF and LF ($P > 0.05$; Figure 5A). Serum glucagon concentrations increased markedly during the first 12 h in both FF and LF ($P < 0.05$) and were on average greater in LF compared with FF until d 4 p.p. ($P < 0.05$; Figure 5B). Cortisol concentrations in plasma continuously decreased in both groups until d 21 p.p. (Figure 5C) and were lower in FF compared with LF during the first 4 d of life ($P = 0.02$; Figure 5C). Concentrations of T_3 and T_4 in plasma were greatest immediately after birth and declined steadily throughout the observation period

(data not shown). Treatment had no effect on thyroid hormone concentrations ($P > 0.05$).

Activities of Aspartate Aminotransferase and Gamma-Glutamyl Transferase

In both FF and LF, a steep increase in ASAT activity was observed, reaching a maximum at 24 h p.p. After d 2 p.p., ASAT activity declined and remained constant until the end of the study. No differences were observed between FF and LF ($P > 0.05$; data not shown). A notable rise in GGT activity was observed in FF and LF after colostrum intake until 12 h p.p. ($P < 0.05$; Figure 4C). In contrast to FF, GGT activity further increased in LF and

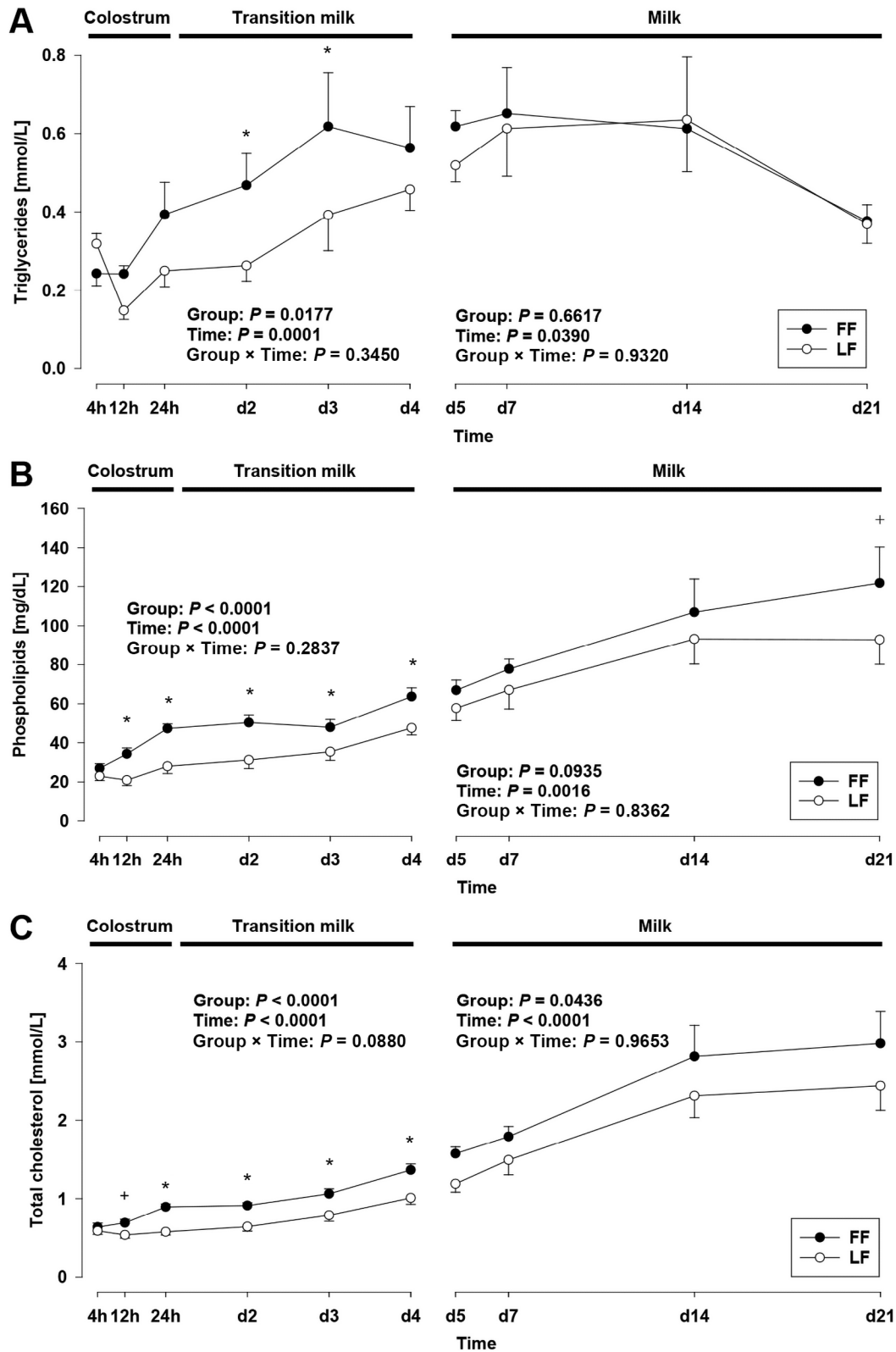


Figure 3. Concentrations of triglycerides (A), phospholipids (B), total cholesterol (C), and high-density lipoprotein cholesterol (HDL-C, D) concentrations in blood of calves during the experiment. The observation periods for FF ($n = 6$) and LF ($n = 6$) were divided into a colostrum phase (4 to 24 h), a transition milk phase (d 2 to 4), and a bulk tank milk phase (d 5–21). Data are presented as mean values $\pm$ SEM. Significant differences ($P < 0.05$) between groups are indicated by *, and tendencies toward significance ($0.05 \leq P < 0.10$) are marked by +.

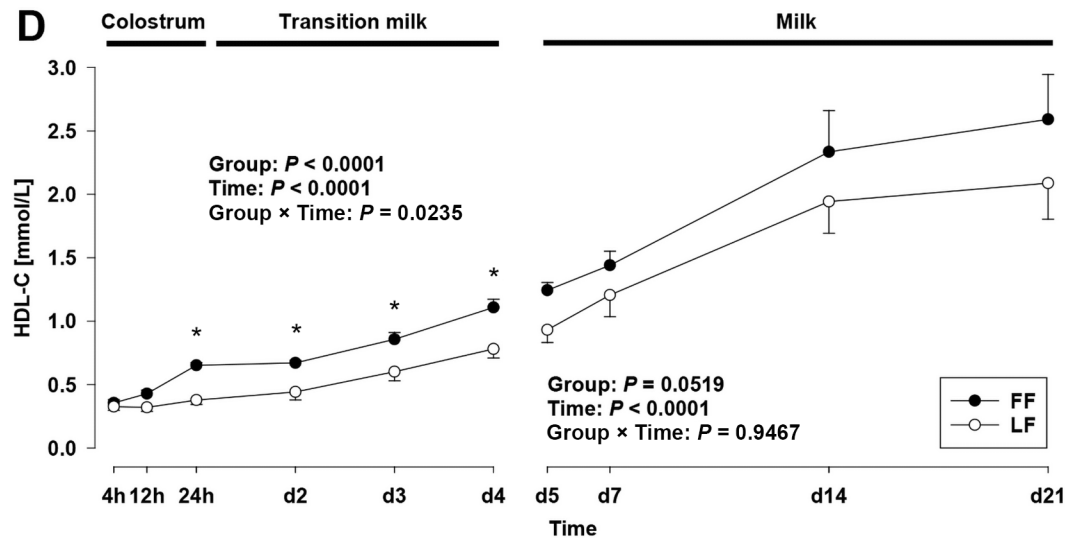


Figure 3 (Continued). Concentrations of triglycerides (A), phospholipids (B), total cholesterol (C), and high-density lipoprotein cholesterol (HDL-C, D) concentrations in blood of calves during the experiment. The observation periods for FF ($n = 6$) and LF ($n = 6$) were divided into a colostrum phase (4 to 24 h), a transition milk phase (d 2 to 4), and a bulk tank milk phase (d 5–21). Data are presented as mean values $\pm$ SEM. Significant differences ($P < 0.05$) between groups are indicated by *, and tendencies toward significance ($0.05 \leq P < 0.10$) are marked by +.

was greater in LF compared with FF during the entire study period ($P < 0.05$; Figure 4C).

DISCUSSION

The present study investigated the impact of differential fat contents in colostrum and transition milk on the metabolic adaptation and development of neonatal calves during the first 3 wk of life during cold ambient temperatures. The average temperatures measured during the experimental period were within the range of observations of previous years and representative of a regular cold season at the experimental station in Switzerland.

Impact of Colostral Fat Content and Cold Environment on Glucose Metabolism

Undoubtedly, thermoregulation in neonatal calves is essential to maintain body temperature and to adapt to extrauterine environmental conditions (Nonnecke et al., 2009; Hammon et al., 2012; Silva et al., 2021). During the cold season, energy expenditure for temperature homeostasis is elevated (Silva et al., 2021). Independent of treatments, we observed a continuous increase in plasma glucose concentration until 24 h p.p. in both groups. This finding is in line with similar results reported by Kurz and Willett (1991) and Bittrich et al. (2002). After colostrum intake, glucose concentrations on d 1 and 2 were greater in FF compared with LF due to the different fat supply via colostrum and transition milk.

Similar to our findings, Hammon and Blum (1998) observed greater glucose and plasma glucagon concentrations in calves fed a diet high in fat but low in carbohydrates. Elevated glucagon concentrations after colostrum intake are consistent with previous research in humans and animal models, as evidenced by studies conducted by Lepine et al. (1991), Ferré et al. (1986), and Girard (1986). In the present study, calves in the LF group had higher average glucagon concentrations than those in the FF group, especially in the first 4 d. Notably, during the first 2 d of life, plasma glucose concentrations in FF calves were on average more than 20% greater than in LF calves. Although gluconeogenesis was not measured in the present study and urea concentrations did not differ among treatments, mobilized AA from protein breakdown are likely to be the most relevant substrates for gluconeogenesis in calves at this age (Aschenbach et al., 2010; Hammon et al., 2013). In contrast, the higher colostrum fat content in the FF group provided additional energy, enabling these calves to achieve markedly greater plasma glucose concentrations despite similar environmental challenges. We assume that the overall greater energy intake in FF spared glucose by prioritized oxidation of fat. The determination of glucose and fat oxidation would be desirable to prove this hypothesis but was out of scope of the present study.

The temporal pattern of glucagon concentrations in both groups showed a similar increase up to at least 12 h p.p. followed by a decline from 24 h onwards, confirming earlier reports from Bittrich et al. (2002) and Hammon and Blum (1998). Although not significant, insulin

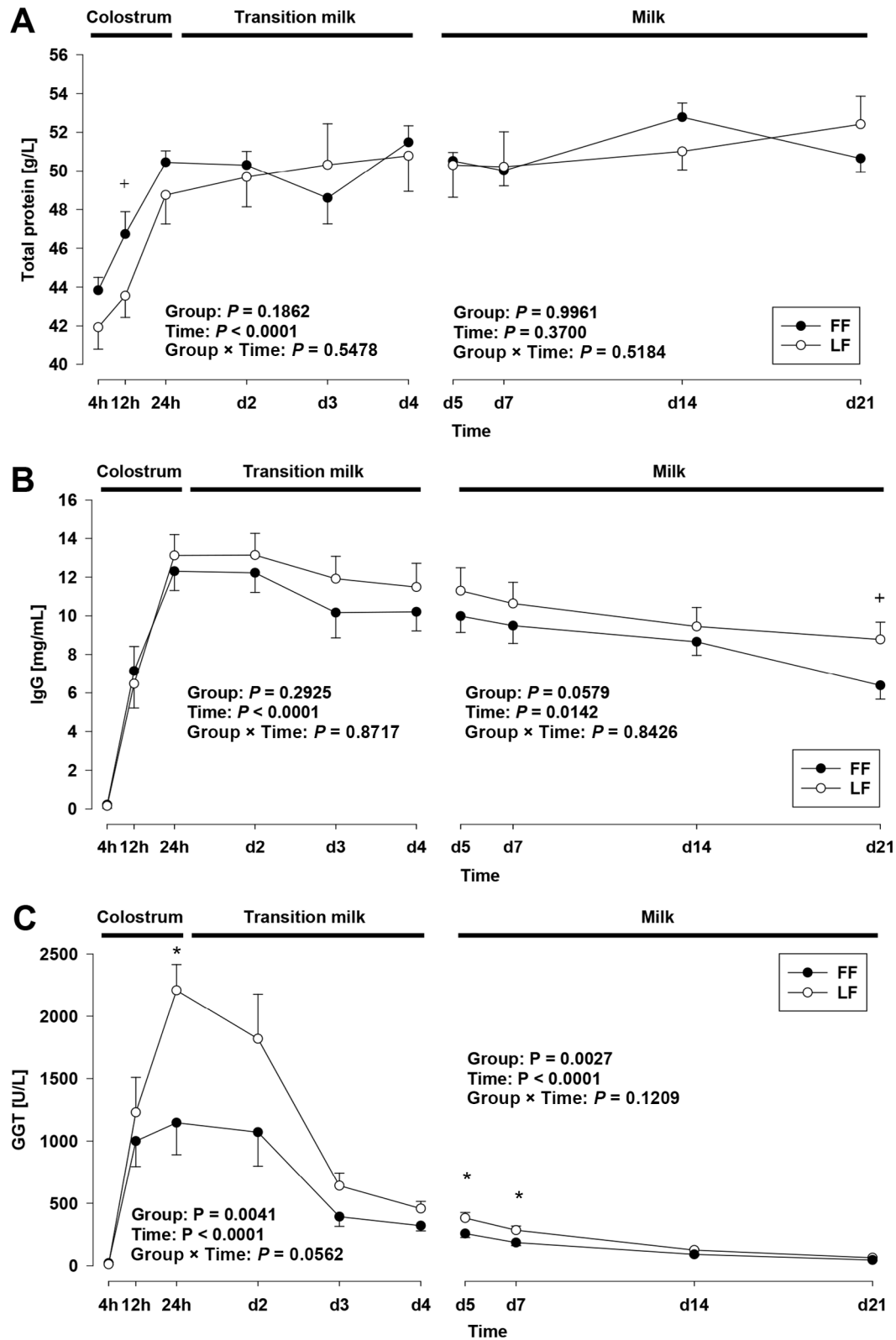


Figure 4. Concentrations of total protein (TP, A), IgG (B), and activity of gamma-glutamyl transferase (GGT, C) in blood of calves during the experiment. The observation periods for FF ($n = 6$) and LF ($n = 6$) were divided into a colostrum phase (4 h to 24 h), a transition milk phase (d 2 to d 4), and a bulk tank milk phase (d 5–21). Data are presented as mean values $\pm$ SEM. Significant differences ($P < 0.05$) between groups are indicated by *, and tendencies toward significance ($0.05 \leq P < 0.10$) are marked by +.

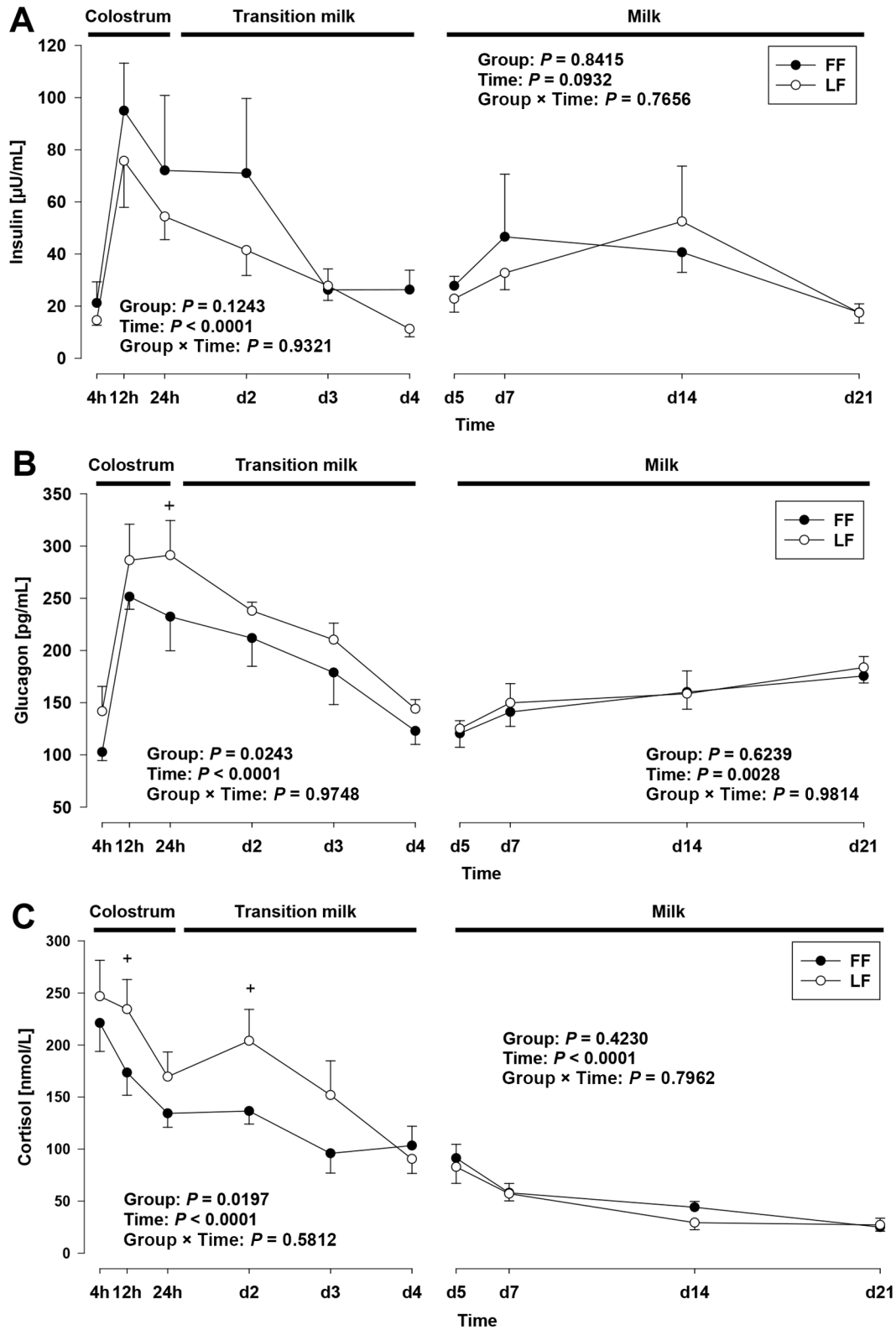


Figure 5. Concentrations of insulin (A), glucagon (B), and cortisol (C) in blood of calves during the experiment. The observation periods for FF ($n = 6$) and LF ($n = 6$) were divided into a colostrum phase (4 to 24 h), a transition milk phase (d 2 to 4), and a bulk tank milk phase (d 5–21). Data are presented as mean values $\pm$ SEM. Tendencies toward significance ($0.05 \leq P < 0.10$) are marked by +.

concentrations appear to be lower in LF calves, which could indicate a reduced need to remove abundant nutrients from circulation.

In addition to glucagon and insulin, cortisol contributes to the regulation of glucose homeostasis during the first 24 h p.p. (Hammon and Blum, 1998). A higher dietary fat content likely spares glucose while concurrently reducing cortisol concentrations. This observation is supported by recent data from von Riedheim et al. (2026) showing that an increasing fat content in colostrum was associated with lower cortisol concentrations in calves. It must be emphasized that cortisol does not contribute to the stimulation of gluconeogenesis but rather supports the development of peripheral insulin resistance (Sternbauer et al., 1998; Scheuer et al., 2006; Kusenda et al., 2013). Consequently, elevated glucose concentrations as observed in FF in the present study may originate partly from less insulin sensitive peripheral tissues. The overall decline in cortisol concentrations in calves during the first week of life aligns with previous findings (Kurz and Willett, 1991; Hadorn et al., 1997; Hammon and Blum, 1998). The elevated cortisol concentrations at birth may be primarily attributed to the calving process and birth-related stress responses, and possibly also to postpartum handling (Stanko et al., 1991; Kovács et al., 2021). Lepine et al. (1989) demonstrated that colostrum-deprived piglets exhibit higher cortisol concentrations than animals with adequate colostrum intake. It remains to be elucidated to which extent cortisol levels are directly influenced by the aforementioned stressors described by Stanko et al. (1991) or modulated by the fat and energy content of colostrum and transition milk.

Plasma concentrations of T_3 and T_4 decreased throughout the entire experimental period, in agreement with earlier studies (Vermorel et al., 1989; Hadorn et al., 1997; Bittrich et al., 2002). Additionally, Grongnet et al. (1985) and Kinsbergen et al. (1994) described declining T_3 and T_4 levels during phases of energy deficit in calves. However, no differences in T_3 and T_4 were observed between LF and FF calves in the present study. Therefore, a decline in T_3 and T_4 concentrations during the first days of life cannot exclusively be attributed to an energy deficit.

Impact of Colostral Fat Content and Cold Environment on Lipid Metabolism

Neonatal calves have only limited body fat reserves that can be mobilized during cold exposure (Roland et al., 2016; Klopp et al., 2022). Immediately after birth and before the first feed intake, NEFA concentrations were greatest in accordance with earlier studies (Vermorel et al., 1989; Hadorn et al., 1997; Rauprich et al., 2000). In both FF and LF, NEFA concentrations declined rapidly within the first 12h p.p. following colostrum feeding

and thus energy intake. Although we did not detect differences between LF and FF, NEFA concentrations are indicative of fat tissue mobilization and their oxidation.

Serum TG closely reflects dietary fat absorption (Leat, 1967; Blum et al., 1997; Arfuso et al., 2025). Elevated TG concentrations in FF calves of the present study were likely due to greater fat intake during the colostrum and transition milk phase. The increasing TG concentrations after colostrum feeding were consistently described in studies by Hammon and Blum (1998) and Rauprich et al. (2000). In addition to greater fat content, bioactive components in full-fat colostrum triggering digestive processes, gut permeability and differentiation, may further account for greater fat absorption. The greater PL and lipoprotein concentrations in FF up to d 4 p.p. further support our findings of a significantly increased fat absorption in FF compared with LF. Similar results were published by Hammon and Blum (1998) and Arfuso et al. (2017). In agreement with Bazin and Brisson (1976), the PL profiles were comparable to those of TG. Concentrations of PL, TC, and HDL-C followed a similar pattern. Arfuso et al. (2025) observed a gradual rise in cholesterol levels up to d 60 p.p. HDL-C constitutes the predominant fraction of cholesterol-associated lipoproteins in calves (Forte et al., 1981; Bauchart et al., 1989; Bauchart, 1993). Serum concentrations of VLDL-C closely resembled those of TG, with both the FF and LF groups showing an increase up to d 4 p.p.

Impact of Colostral Fat Content and Cold Environment on Serum Enzyme Activities, IgG, and TP Concentrations

The importance of IgG uptake and its interaction with GGT and TP has been well established across various species (Pauli, 1983; Center et al., 1991; Weaver et al., 2000). In the present study, serum IgG, TP, and GGT demonstrated a marked increase in concentration or activity levels, or both, by 24 h p.p. in agreement with Perino et al. (1993) and Bittrich et al. (2002), who confirmed positive correlations between GGT and TP with IgG. Due to the similar IgG content in FF and LF colostrum, no differences were observed between FF and LF for serum TP and IgG concentrations. The postnatal increase in ASAT activity has been described earlier by Kurz and Willett (1991) and Zanker et al. (2001). In the absence of a significant effect over time for FF and LF, we assume that colostrum fat content does not affect ASAT activity during the initial days of life. Consequently, elevated activity values can be attributed to its endogenous production (Hammon and Blum, 1998; Zanker et al., 2001). In contrast, GGT activity was higher in LF than in FF during the first 4 d p.p. This finding aligns with observations reported by von Riedheim et al. (2026), who noted

a negative correlation between colostrum fat content and serum GGT activity in calves. We can only speculate if elevated GGT activity indicates a generally increased hepatic activity of concomitant metabolic processes in LF.

Limitations of the Study

In addition to the relatively low number of enrolled calves, we assume that during colder temperature exposure, the metabolic and endocrine differences would be more pronounced among FF and LF. Nevertheless, under the relatively moderate cold environment of the present study and despite the low number of animals, we see a clear impact of colostrum fat content on metabolic adaptation in neonatal calves.

CONCLUSIONS

Our findings highlight that an adequate colostrum fat content plays a crucial role in supporting metabolic stability in calves exposed to cold temperatures. Neonatal calves fed colostrum and milk with a markedly reduced fat content exhibit distinctly lower plasma glucose concentrations under cold environmental conditions. In contrast, feeding a full-fat colostrum was accompanied by lower plasma cortisol and glucagon concentrations and improved glucose homeostasis. Practically, this suggests that colostrum management and quality assessment should consider not only Ig concentrations but also the energy and fat content of the first meals. Particularly in herds with winter calvings, providing fat-rich colostrum may help prevent early-life energy deficits, reduce susceptibility to disease, and support optimal growth.

NOTES

The authors gratefully acknowledge the support of Phytobiotics Futterzusatzstoffe GmbH (Eltville, Germany). The support of the laboratory staff of the Veterinary Physiology group (Bern, Switzerland) is gratefully acknowledged. The present study was carried out in accordance with the Swiss animal protection and welfare legislation and approved by the cantonal animal experimentation commission of the Canton of Fribourg (Switzerland, approval number 2024-23-FR). The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: ASAT = aspartate aminotransferase; CS = compound symmetry; FF = full-fat; GGT = gamma-glutamyl transferase; GLU = plasma glucose; HDL-C = high-density lipoprotein cholesterol; LDL-C = low-density lipoprotein cholesterol; LF = low-fat; max. = maximum; min. = minimum; ND = not determined; NEFA = nonesterified fatty acids; PL

= phospholipid; p.p. = postpartum; ref. = reference; T₃ = triiodothyronine; T₄ = thyroxine; TC = total cholesterol; TG = triglyceride; TP = total protein; VC = variance components; VLDL-C = very-low-density lipoprotein cholesterol.

REFERENCES

- Arfuso, F., F. Arrigo, V. Lopreiato, M. Rizzo, L. Liotta, C. Giannetto, and G. Piccione. 2025. Changes in plasma UCP-1, leptin, lipids, and pro-inflammatory interleukins in calves from birth to weaning. *Res. Vet. Sci.* 189:105627. <https://doi.org/10.1016/j.rvsc.2025.105627>.
- Arfuso, F., F. Fazio, M. Panzera, C. Giannetto, S. Di Pietro, E. Giudice, and G. Piccione. 2017. Lipid and lipoprotein profile changes in newborn calves in response to the perinatal period. *Acta Vet. (Beogr.)* 67:25–32. <https://doi.org/10.1515/aeve-2017-0003>.
- Aschenbach, J. R., N. B. Kristensen, S. S. Donkin, H. M. Hammon, and G. B. Penner. 2010. Gluconeogenesis in dairy cows: The secret of making sweet milk from sour dough. *IUBMB Life* 62:869–877. <https://doi.org/10.1002/iub.400>.
- Bauchart, D. 1993. Lipid absorption and transport in ruminants. *J. Dairy Sci.* 76:3864–3881. [https://doi.org/10.3168/jds.S0022-0302\(93\)77728-0](https://doi.org/10.3168/jds.S0022-0302(93)77728-0).
- Bauchart, D., D. Durand, P. M. Laplaud, P. Forgez, S. Goulinet, and M. J. Chapman. 1989. Plasma lipoproteins and apolipoproteins in the pre-ruminant calf, *Bos* spp: Density distribution, physicochemical properties, and the in vivo evaluation of the contribution of the liver to lipoprotein homeostasis. *J. Lipid Res.* 30:1499–1514. [https://doi.org/10.1016/S0022-2275\(20\)38235-3](https://doi.org/10.1016/S0022-2275(20)38235-3).
- Bazin, R. C., and G. J. Brisson. 1976. Plasma lipids, ketone bodies, and glucose concentrations in calves fed high-and low-fat milk replacers. *J. Dairy Sci.* 59:1301–1305. [https://doi.org/10.3168/jds.S0022-0302\(76\)84359-7](https://doi.org/10.3168/jds.S0022-0302(76)84359-7).
- Bienboire-Frosini, C., D. Wang, M. Marcet-Rius, D. Villanueva-García, A. Gazzano, A. Domínguez-Oliva, A. Olmos-Hernández, I. Hernández-Ávalos, K. Lezama-García, A. Verduzco-Mendoza, J. Gómez-Prado, and D. Mota-Rojas. 2023. The role of brown adipose tissue and energy metabolism in mammalian thermoregulation during the perinatal period. *Animals (Basel)* 13:2173. <https://doi.org/10.3390/ani13132173>.
- Bigler, N. A., R. M. Bruckmaier, and J. J. Gross. 2022. Implications of placental type on species-specific colostrum properties in mammals. *J. Anim. Sci.* 100:skac287. <https://doi.org/10.1093/jas/skac287>.
- Bittrich, S., C. Morel, C. Philipona, Y. Zbinden, H. M. Hammon, and J. W. Blum. 2002. Physiological traits in preterm calves during their first week of life. *J. Anim. Physiol. Anim. Nutr. (Berl.)* 86:185–198. <https://doi.org/10.1046/j.1439-0396.2002.00372.x>.
- Blum, J. W., U. Hadorn, H. P. Sallmann, and W. Schuep. 1997. Delaying colostrum intake by one day impairs plasma lipid, essential fatty acid, carotene, retinol and alpha-tocopherol status in neonatal calves. *J. Nutr.* 127:2024–2029. <https://doi.org/10.1093/jn/127.10.2024>.
- Center, S. A., J. F. Randolph, T. ManWarren, and M. Slater. 1991. Effect of colostrum ingestion on gamma-glutamyltransferase and alkaline phosphatase activities in neonatal pups. *Am. J. Vet. Res.* 52:499–504. <https://doi.org/10.2460/ajvr.1991.52.03.499>.
- Ferré, P., J. F. Decaux, T. Issad, and J. Girard. 1986. Changes in energy metabolism during the suckling and weaning period in the newborn. *Reprod. Nutr. Dev.* 26(2B):619–631. <https://doi.org/10.1051/rnd:19860413>.
- Forte, T. M., J. J. Bell-Quint, and F. Cheng. 1981. Lipoproteins of fetal and newborn calves and adult steer: A study of developmental changes. *Lipids* 16:240–245. <https://doi.org/10.1007/BF02535023>.
- Friedewald, W. T., R. I. Levy, and D. S. Fredrickson. 1972. Estimation of the concentration of low-density lipoprotein cholesterol in plasma, without use of the preparative ultracentrifuge. *Clin. Chem.* 18:499–502. <https://doi.org/10.1093/clinchem/18.6.499>.
- Girard, J. 1986. Gluconeogenesis in late fetal and early neonatal life. *Neonatology* 50:237–258. <https://doi.org/10.1159/000242605>.

- Grongnet, J. F., J. E. Grongnet-Pinchon, A. Witowski, D. Chevreil, R. Garnier, J. Lareynie, and M. Lesage. 1985. Neonatal levels of plasma thyroxine in male and female calves fed a colostrum or immunoglobulin diet or fasted for the first 28 hours of life. *Reprod. Nutr. Dev.* 25:537–543. <https://doi.org/10.1051/rnd:19850406>.
- Hadorn, U., H. Hammon, R. M. Bruckmaier, and J. W. Blum. 1997. Delaying colostrum intake by one day has important effects on metabolic traits and on gastrointestinal and metabolic hormones in neonatal calves. *J. Nutr.* 127:2011–2023. <https://doi.org/10.1093/jn/127.10.2011>.
- Hammon, H. M., and J. W. Blum. 1998. Metabolic and endocrine traits of neonatal calves are influenced by feeding colostrum for different durations or only milk replacer. *J. Nutr.* 128:624–632. <https://doi.org/10.1093/jn/128.3.624>.
- Hammon, H. M., J. Steinhoff-Wagner, J. Flor, U. Schönhusen, and C. C. Metges. 2013. Lactation Biology Symposium: Role of colostrum and colostrum components on glucose metabolism in neonatal calves. *J. Anim. Sci.* 91:685–695. <https://doi.org/10.2527/jas.2012-5758>.
- Hammon, H. M., J. Steinhoff-Wagner, U. Schönhusen, C. C. Metges, and J. W. Blum. 2012. Energy metabolism in the newborn farm animal with emphasis on the calf: Endocrine changes and responses to milk-born and systemic hormones. *Domest. Anim. Endocrinol.* 43:171–185. <https://doi.org/10.1016/j.domaniend.2012.02.005>.
- Kertz, A. F., and H. Chester-Jones. 2004. Invited Review: Guidelines for measuring and reporting calf and heifer experimental data. *J. Dairy Sci.* 87:3577–3580. [https://doi.org/10.3168/jds.S0022-0302\(04\)73495-5](https://doi.org/10.3168/jds.S0022-0302(04)73495-5).
- Kessler, E. C., J. J. Gross, R. M. Bruckmaier, and C. Albrecht. 2014. Cholesterol metabolism, transport, and hepatic regulation in dairy cows during transition and early lactation. *J. Dairy Sci.* 97:5481–5490. <https://doi.org/10.3168/jds.2014-7926>.
- Kinsbergen, M., H.-P. Sallmann, and J. W. Blum. 1994. Metabolic, endocrine and haematological changes in 1-week-old calves after milk intake, in response to fasting and during total parenteral nutrition. *J. Vet. Med. A Physiol. Pathol. Clin. Med.* 41:268–282. <https://doi.org/10.1111/j.1439-0442.1994.tb00094.x>.
- Klopp, R. N., C. R. Ferreira, T. M. Casey, and J. P. Boerman. 2022. Relationship of cow and calf circulating lipidomes with colostrum lipid composition and metabolic status of the cow. *J. Dairy Sci.* 105:1768–1787. <https://doi.org/10.3168/jds.2021-21008>.
- Kovács, L., F. L. Kézér, S. Bodó, F. Ruff, R. Palme, and O. Szenci. 2021. Salivary cortisol as a non-invasive approach to assess stress in dystocic dairy calves. *Sci. Rep.* 11:6200. <https://doi.org/10.1038/s41598-021-85666-9>.
- Kurz, M. M., and L. B. Willett. 1991. Carbohydrate, enzyme, and hematology dynamics in newborn calves. *J. Dairy Sci.* 74:2109–2118. [https://doi.org/10.3168/jds.S0022-0302\(91\)78383-5](https://doi.org/10.3168/jds.S0022-0302(91)78383-5).
- Kusenda, M., M. Kaske, M. Piechotta, L. Locher, A. Starke, K. Huber, and J. Rehage. 2013. Effects of dexamethasone-21-isonicotinate on peripheral insulin action in dairy cows 5 days after surgical correction of abomasal displacement. *J. Vet. Intern. Med.* 27:200–206. <https://doi.org/10.1111/jvim.12010>.
- Leat, W. M. F. 1967. Plasma lipids of newborn and adult ruminants and of lambs from birth to weaning. *J. Agric. Sci.* 69:241–246. <https://doi.org/10.1017/S0021859600018645>.
- Lepine, A. J., R. D. Boyd, and J. Welch. 1989. Effect of colostrum intake on plasma glucose, non-esterified fatty acid and glucoregulatory hormone patterns in the neonatal pig. *Domest. Anim. Endocrinol.* 6:231–241. [https://doi.org/10.1016/0739-7240\(89\)90017-9](https://doi.org/10.1016/0739-7240(89)90017-9).
- Lepine, A. J., R. D. Boyd, and D. M. Whitehead. 1991. Effect of colostrum intake on hepatic gluconeogenesis and fatty acid oxidation in the neonatal pig. *J. Anim. Sci.* 69:1966–1974. <https://doi.org/10.2527/1991.6951966x>.
- Litherland, N. B., D. N. Da Silva, R. J. LaBerge, J. Schefers, and A. Kertz. 2014. Supplemental fat for dairy calves during mild cold stress. *J. Dairy Sci.* 97:2980–2989. <https://doi.org/10.3168/jds.2013-6942>.
- McGuirk, S. M. 2008. Disease management of dairy calves and heifers. *Vet. Clin. North Am. Food Anim. Pract.* 24:139–153. <https://doi.org/10.1016/j.cvfa.2007.10.003>.
- McGuirk, S. M., and S. F. Peek. 2014. Timely diagnosis of dairy calf respiratory disease using a standardized scoring system. *Anim. Health Res. Rev.* 15:145–147. <https://doi.org/10.1017/S1466252314000267>.
- Nonnecke, B. J., M. R. Foote, B. L. Miller, M. Fowler, T. E. Johnson, and R. L. Horst. 2009. Effects of chronic environmental cold on growth, health, and select metabolic and immunologic responses of preruminant calves. *J. Dairy Sci.* 92:6134–6143. <https://doi.org/10.3168/jds.2009-2517>.
- Odle, J. 1997. New insights into the utilization of medium-chain triglycerides by the neonate: Observations from a piglet model. *J. Nutr.* 127:1061–1067. <https://doi.org/10.1093/jn/127.6.1061>.
- Pauli, J. V. 1983. Colostral transfer of gamma glutamyl transferase in lambs. *N. Z. Vet. J.* 31:150–151. <https://doi.org/10.1080/00480169.1983.35004>.
- Perino, L. J., R. L. Sutherland, and N. E. Woollen. 1993. Serum gamma-glutamyltransferase activity and protein concentration at birth and after suckling in calves with adequate and inadequate passive transfer of immunoglobulin G. *Am. J. Vet. Res.* 54:56–59. <https://doi.org/10.2460/ajvr.1993.54.01.56>.
- Polk, D. H. 1988. Thyroid hormone effects on neonatal thermogenesis. *Semin. Perinatol.* 12:151–156.
- Rauprich, A. B., H. M. Hammon, and J. W. Blum. 2000. Effects of feeding colostrum and a formula with nutrient contents as colostrum on metabolic and endocrine traits in neonatal calves. *Neonatology* 78:53–64. <https://doi.org/10.1159/000014247>.
- Roland, L., M. Drillich, D. Klein-Jöbstl, and M. Iwersen. 2016. Invited review: Influence of climatic conditions on the development, performance, and health of calves. *J. Dairy Sci.* 99:2438–2452. <https://doi.org/10.3168/jds.2015-9901>.
- Scheuer, B. H., Y. Zbinden, P. Schneiter, L. Tappy, J. W. Blum, and H. M. Hammon. 2006. Effects of colostrum feeding and glucocorticoid administration on insulin-dependent glucose metabolism in neonatal calves. *Domest. Anim. Endocrinol.* 31:227–245. <https://doi.org/10.1016/j.domaniend.2005.11.004>.
- Silva, F. L. M., E. Miqueo, M. D. da Silva, T. M. Torrezan, N. B. Rocha, M. S. V. Salles, and C. M. M. Bittar. 2021. Thermoregulatory responses and performance of dairy calves fed different amounts of colostrum. *Animals (Basel)* 11:703. <https://doi.org/10.3390/ani11030703>.
- Sjaunja, L. O., L. Baevre, L. Junkkarinen, J. Pedersen, and J. Setälä. 1990. Measurement of the total energy content of cow's milk and the energy value of milk fat and milk protein. Pages 152–155 in 26th Session of the International Committee for Recording the Productivity of Milk Animals (ICRPMA). Oslo, Norway. Pudoc, Wageningen, the Netherlands.
- Stanko, R. L., M. J. Guthrie, and R. D. Randel. 1991. Response to environmental temperatures in Brahman calves during the first compared to the second day after birth. *J. Anim. Sci.* 69:4419–4427. <https://doi.org/10.2527/1991.69114419x>.
- Sternbauer, K., J. Luthman, A. Hänni, and S. O. Jacobsson. 1998. Clenbuterol-induced insulin resistance in calves measured by hyperinsulinemic, euglycemic clamp technique. *Acta Vet. Scand.* 39:281–289. <https://doi.org/10.1186/BF03547800>.
- Su, D., T. Zhou, Y. Wang, and L. Wang. 2023. Cold exposure regulates hepatic glycogen and lipid metabolism in newborn goats. *Int. J. Mol. Sci.* 24:14330. <https://doi.org/10.3390/ijms241814330>.
- Vermorel, M., J. Vernet, Saido, C. Dardillat, C. Demigne, and M.-J. Davicco. 1989. Energy metabolism and thermoregulation in the newborn calf; effect of calving conditions. *Can. J. Anim. Sci.* 69:113–122. <https://doi.org/10.4141/cjas89-014>.
- von Riedheim, M. A. E., E. C. Kessler, and J. J. Gross. 2026. Associations of individual nutrient and non-nutrient factors on endocrine and metabolic parameters in neonatal calves fed colostrum from their own dams. *J. Dairy Sci.* 109:3048–3062. <https://doi.org/10.3168/jds.2025-27341>.
- Weaver, D. M., J. W. Tyler, D. C. VanMetre, D. E. Hostetler, and G. M. Barrington. 2000. Passive transfer of colostral immunoglobulins in calves. *J. Vet. Intern. Med.* 14:569–577. <https://doi.org/10.1111/j.1939-1676.2000.tb02278.x>.

- Young, B. A. 1981. Cold stress as it affects animal production. *J. Anim. Sci.* 52:154–163. <https://doi.org/10.2527/jas1981.521154x>.
- Zanker, I. A., H. M. Hammon, and J. W. Blum. 2001. Activities of gamma-glutamyltransferase, alkaline phosphatase and aspartate-aminotransferase in colostrum, milk and blood plasma of calves fed first colostrum at 0–2, 6–7, 12–13 and 24–25 h after birth. *J. Vet. Med. A Physiol. Pathol. Clin. Med.* 48:179–185. <https://doi.org/10.1046/j.1439-0442.2001.00338.x>.

ORCIDS

- Hannes Erkinger,  <https://orcid.org/0009-0006-0068-1661>
- Maximilian A. E. von Riedheim,  <https://orcid.org/0009-0007-3022-3722>
- Oguz Calisici,  <https://orcid.org/0000-0001-9317-7486>
- Josef J. Gross  <https://orcid.org/0000-0002-2578-6076>